

Question number	Answer	Additional guidance	
4(a)	An answer that includes four of the following points: • neurotransmitters bind to (post- synaptic) membrane (1) • sodium ion channels open (1) • sodium ions diffuse in (to the axon) (1) • (positive feedback results in) more sodium ion channels open (1) • resulting in depolarisation (of the membrane) (1) • correct reference to threshold (potential) (1)	ACCEPT Na⁺ for sodium ions ACCEPT sodium ions enter {axon / neurone} ACCEPT depolarisation described eg voltage to +40mV IGNORE depolarisation of axon	(4)

Question number	Answer	Additional guidance	Mark
4(b)	A calculation showing the following steps: • calculation of gradient at steepest part of upward curve (1) • answer given to two significant figures with correct units (1)	ECF for mp 2 providing dy/dx calculated and given to 2 significant figures $(9 \div 0.5)$ or 18. = 18 a.u. millisecond⁻¹ correct answer scores full marks accept range 17.0- 20.0 a.u. millisecond⁻¹	(2)

Question number	Answer	Additional guidance	Mark
4(c)	A is the correct answer B is not the correct answer as 0.3:1 is not the correct ratio C is not the correct answer as 0.083:1 is not the correct ratio D is not the correct answer as 1:38.75 is not the correct ratio		(1)

Question number	Answer	Additional guidance	Mark
4(d)(i)	An answer that includes these two the points - increased / faster / speed up (nerve impulse) (1) - ion channels open (and close) more quickly / (sodium) ions diffuse more quickly / (sodium) ions have more kinetic energy (1)	ACCEPT Na⁺ for sodium ions ignore particles ACCEPT increased rate of transmissions IGNORE enzyme activity ACCEPT ions move quicker / more influx of ions IGNORE increased permeability to ions for 2 marks accept decreased speed due to denaturation of ion channels ONLY	(2)

Question number	Answer	Additional guidance	Mark
4(d)(ii)	A description that includes two of the following points: • the larger the diameter the higher the speed (of impulse) (1) • myelination increases speed (of impulse) / absence of myelin decreases speed (of impulse) (1) • myelination has greater effect than axon diameter	IGNORE if answers comparing neurones A/B/C/D without reference to myelination / axon diameter ACCEPT converse	(2)